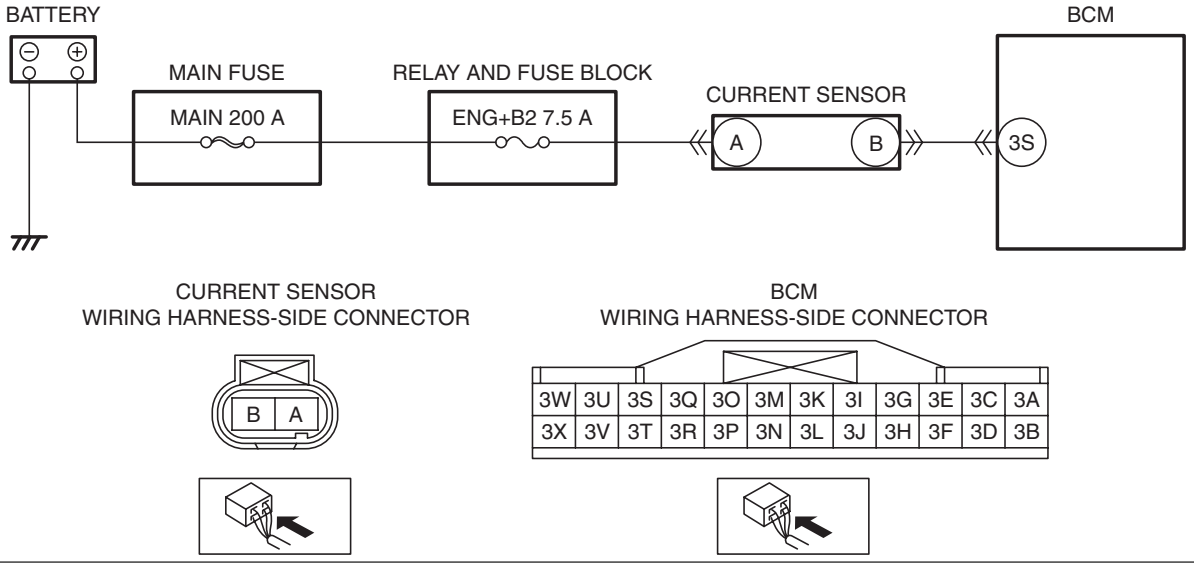
DTC U1007:68 [BCM]

id0902f5071700

Without i-ELOOP

Description	Communication error with current sensor
Detection condition	• BCM could not receive signals from the current sensor for **5 s or more** with the ignition switched ON (engine off or on).
Fail-safe	Not applicable
Possible cause	• Current sensor connector or terminal malfunction • Current sensor power supply circuit malfunction — Short to ground in wiring harness between MAIN 200 A fuse and current sensor terminal A — ENG+B2 7.5 A fuse malfunction — MAIN 200 A fuse malfunction — Open circuit in wiring harness between battery positive terminal and current sensor terminal A • BCM connector or terminal malfunction • Short to ground in wiring harness between current sensor terminal B and BCM terminal 3S • Open circuit in wiring harness between current sensor terminal B and BCM terminal 3S • Current sensor malfunction • BCM malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY BCM DTCs AGAIN** • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Go to the next step.
		No	Go to Step 9.

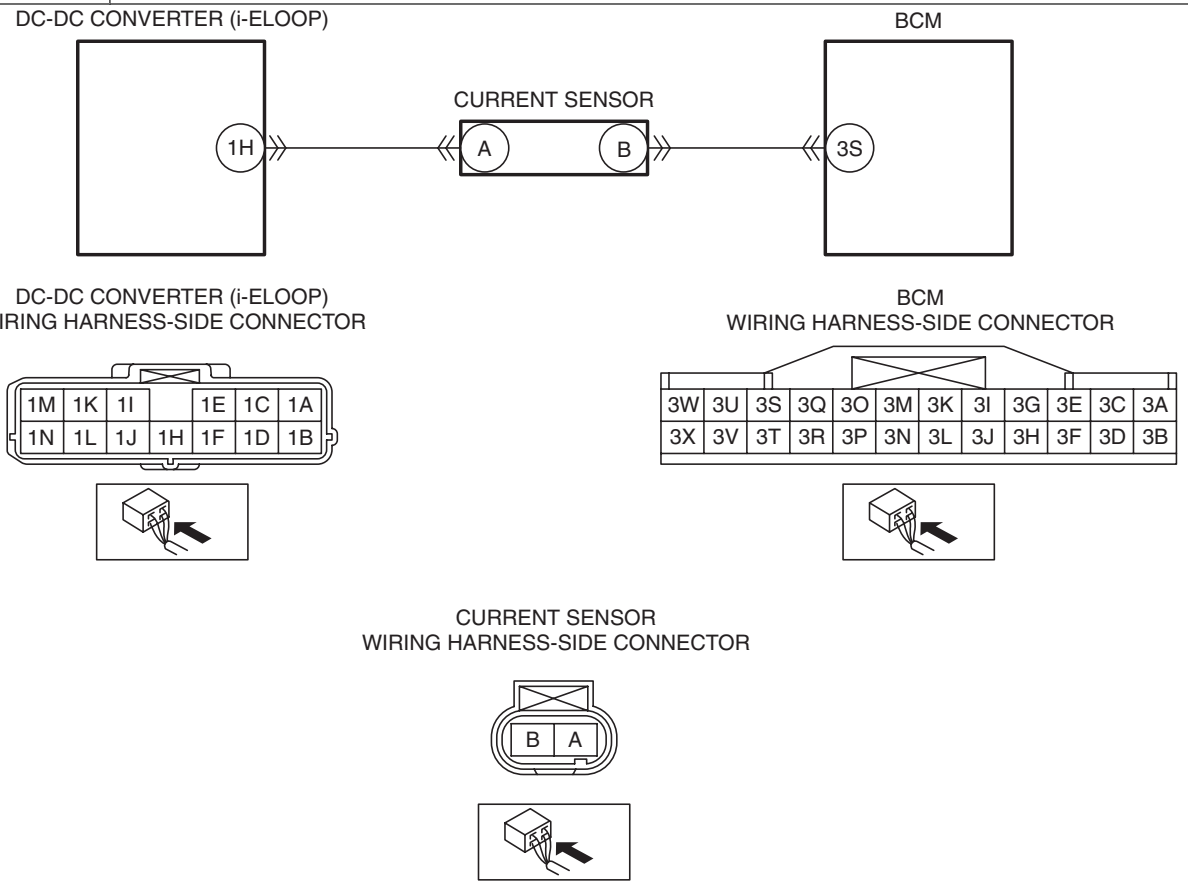
Step	Inspection		Action
2	**INSPECT CURRENT SENSOR CONNECTOR CONDITION** **Note** • Always disconnect current sensor connector before disconnecting the negative battery cable. • Switch the ignition off. • Disconnect the current sensor connector. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 8.
3	**INSPECT CURRENT SENSOR POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT OR SHORT TO GROUND** • Verify that the current sensor connector is disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at the current sensor terminal A (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the ENG+B2 7.5 A fuse and MAIN 200A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200A fuse and current sensor terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and current sensor terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 8.
4	**INSPECT BCM CONNECTOR CONDITION** • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the BCM connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 8.

Step	Inspection		Action
5	**INSPECT CURRENT SENSOR LIN COMMUNICATION CIRCUIT FOR SHORT TO GROUND** • Verify that the current sensor and BCM connectors are disconnected. • Inspect for continuity between current sensor terminal B (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between current sensor terminal B and BCM terminal 3S. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 8.
		No	Go to the next step.
6	**INSPECT CURRENT SENSOR LIN COMMUNICATION CIRCUIT FOR OPEN CIRCUIT** • Verify that the current sensor and BCM connectors are disconnected. • Inspect the wiring harness for continuity between current sensor terminal B (wiring harness-side) and BCM terminal 3S (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between current sensor terminal B and BCM terminal 3S. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 8.
7	**INSPECT CURRENT SENSOR** • Inspect the current sensor. (See CURRENT SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CURRENT SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is the current sensor normal?	Yes	Go to the next step.
		No	Replace the current sensor, then go to the next step. (See CURRENT SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CURRENT SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
8	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
9	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [BCM].)
		No	DTC troubleshooting completed.

With i-ELOOP

Description	Communication error with current sensor
Detection condition	• BCM could not receive signals from the current sensor for **5 s or more** with the ignition switched ON (engine off or on).
Fail-safe	Not applicable

Description	Communication error with current sensor
Possible cause	• Current sensor connector or terminal malfunction • DC-DC converter (i-ELOOP) connector or terminal malfunction • Short to ground in wiring harness between DC-DC converter (i-ELOOP) terminal 1H and current sensor terminal A • Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 1H and current sensor terminal A • BCM connector or terminal malfunction • Short to ground in wiring harness between current sensor terminal B and BCM terminal 3S • Open circuit in wiring harness between current sensor terminal B and BCM terminal 3S • Current sensor malfunction • DC-DC converter (i-ELOOP) malfunction • BCM malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY BCM DTCs AGAIN** • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Go to the next step.
		No	Go to Step 13.
2	**VERIFY DC-DC CONVERTER (i-ELOOP) DTCs** • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].)
		No	Go to the next step.

Step	Inspection		Action
3	**INSPECT CURRENT SENSOR CONNECTOR CONDITION** **Note** • Always disconnect current sensor connector before disconnecting the negative battery cable. • Switch the ignition off. • Disconnect the current sensor connector. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 12.
4	**INSPECT DC-DC CONVERTER (i-ELOOP) CONNECTOR CONDITION** • Disconnect the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) • Disconnect the DC-DC converter (i-ELOOP) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 12.
5	**INSPECT CURRENT SENSOR POWER SUPPLY CIRCUIT FOR SHORT TO GROUND** • Verify that the current sensor and DC-DC converter (i-ELOOP) connectors are disconnected. • Inspect for continuity between current sensor terminal A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 1H and current sensor terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 12.
		No	Go to the next step.
6	**INSPECT CURRENT SENSOR POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Verify that the current sensor and DC-DC converter (i-ELOOP) connectors are disconnected. • Inspect the wiring harness for continuity between DC-DC converter (i-ELOOP) terminal 1H (wiring harness-side) and current sensor terminal A (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 1H and current sensor terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 12.

Step	Inspection		Action
7	**INSPECT BCM CONNECTOR CONDITION** • Disconnect the BCM connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 12.
8	**INSPECT CURRENT SENSOR LIN COMMUNICATION CIRCUIT FOR SHORT TO GROUND** • Verify that the current sensor and BCM connectors are disconnected. • Inspect for continuity between current sensor terminal B (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between current sensor terminal B and BCM terminal 3S. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 12.
		No	Go to the next step.
9	**INSPECT CURRENT SENSOR LIN COMMUNICATION CIRCUIT FOR OPEN CIRCUIT** • Verify that the current sensor and BCM connectors are disconnected. • Inspect the wiring harness for continuity between current sensor terminal B (wiring harness-side) and BCM terminal 3S (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between current sensor terminal B and BCM terminal 3S. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 12.
10	**INSPECT CURRENT SENSOR** • Inspect the current sensor. (See CURRENT SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CURRENT SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is the current sensor normal?	Yes	Go to the next step.
		No	Replace the current sensor, then go to Step 12. (See CURRENT SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CURRENT SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
11	**INSPECT DC-DC CONVERTER (i-ELOOP)** • Inspect the DC-DC converter (i-ELOOP). (See DC-DC CONVERTER (i-ELOOP) INSPECTION [i-ELOOP].) • Is the DC-DC converter (i-ELOOP) normal?	Yes	Go to the next step.
		No	Replace the DC-DC converter (i-ELOOP), then go to the next step. (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)

Step	Inspection		Action
12	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
13	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [BCM].)
		No	DTC troubleshooting completed.